ECE 493 Software Systems Design Project
Winter 2026
Laboratory #3
Dr. Scott Dick

Due Date: Monday Mar. 2, 2026, 11:59:59pm

Additional Resources

https://nodejs.org/api/test.html

https://www.npmjs.com/package/supertest

https://github.com/bcoe/c8

Discussion
With the planning and task generation steps in Spec-kit completed in Lab #2, it's now time to generate the code and tests. Codex will generate the code as requested, as well as unit and integration tests. You will also instruct it to implement the acceptance test suite that we planned out in Lab #1. During testing, the Instructor observed that Codex was unwilling to implement all features in a single batch (and in truth that might not have been wise anyway). Instead, the Instructor's version implemented features (each one a use case) in the same order as they were planned in Lab #2.

Validating generated code will most sensibly be done via testing, as usual. You are to order Codex to produce unit and integration test cases that cover all branches within the code; all tests must then pass. Code coverage is measured using tools besides Node.js test or Supertest; the `c8` tool in particular was suggested by Codex. One particularly important note: Codex runs in a sandbox, for security; however, that means it cannot run code requiring network calls. You'll have to instruct Codex to run with elevated permissions in order for it to run your tests. Alternatively, you can run the testing and code coverage tools yourself, saving many tokens. These are already automated testing tools; there should be no need to run them from within the Codex shell. (You are in fact encouraged to think about what existing tools can be used to augment the AI tools; we believe there is a place for traditional software tool support even in agentic software development.)

Note that debugging, if needed for the tests to all pass, should take place within Codex to maintain the overall structure of the codebase and its relationship to the plans and tasks developed in Spec-kit.

Requirements
You will submit your completed source tree (which is to include your tests), and a brief report. The report should include:

- The procedures you followed to execute the tests (including prompts if you ran the tests via Codex).
- A status report on tests that pass (after debugging).
- A coverage report that should show 100% branch coverage.

Submission

- Submit your source tree through the Department's git server; Jesse will provide instructions on this step. (This is also how your group project implemetnations will be submitted.)
- Submit your report via the submission link in Canvas

Grading

Working code	20%
All tests pass	50%
100% branch coverage achieved	30%